Lab:4
Title Amplitude Modulation and Demodulation

Objectives:

- Understanding the principle of amplitude modulation (AM) and demodulation
- Understanding the waveforms of modulated and demodulated signals

Theory

Modulation is the process of impressing a low-frequency intelligence signal onto a high frequency carrier signal. Amplitude Modulation (AM) is the process that a high-frequency carrier signal is modulated by a low-frequency modulating signal (usually an audio). In amplitude modulation the carrier amplitude varies with the modulating amplitude, as shown in Fig. 1. If the audio signal (message signal) is $A_m \cos(2\pi f_m t)$ and the carrier signal is $A_c \cos(2\pi f_c t)$, the amplitude-modulated signal can be expressed by

$$x_{AM}(t) = [1 + A_m \cos(2\pi f_m t)]A_c \cos(2\pi f_c t) \tag{1}$$

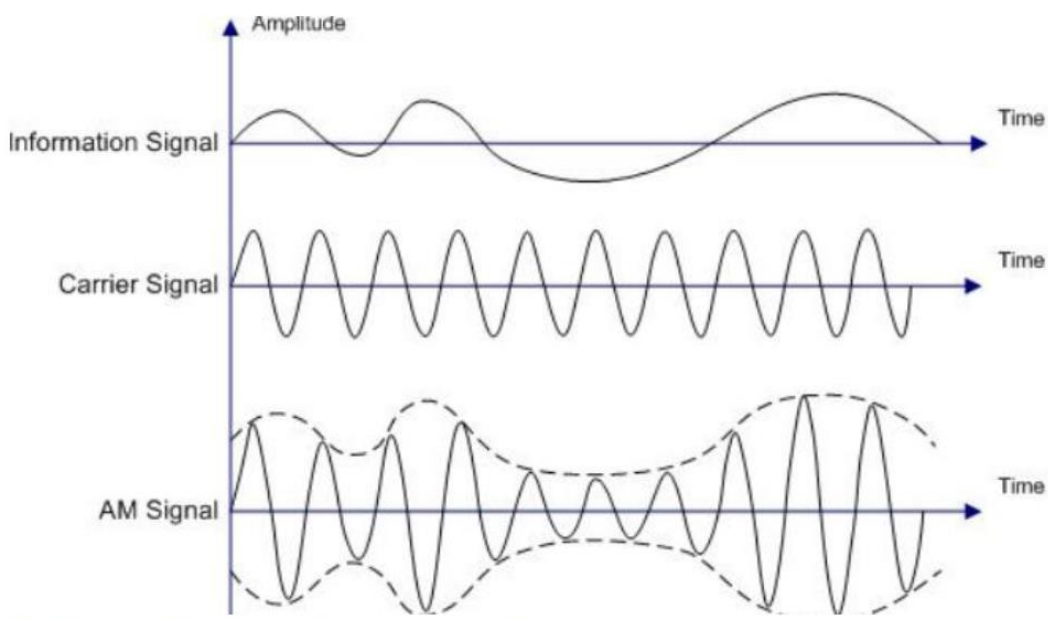
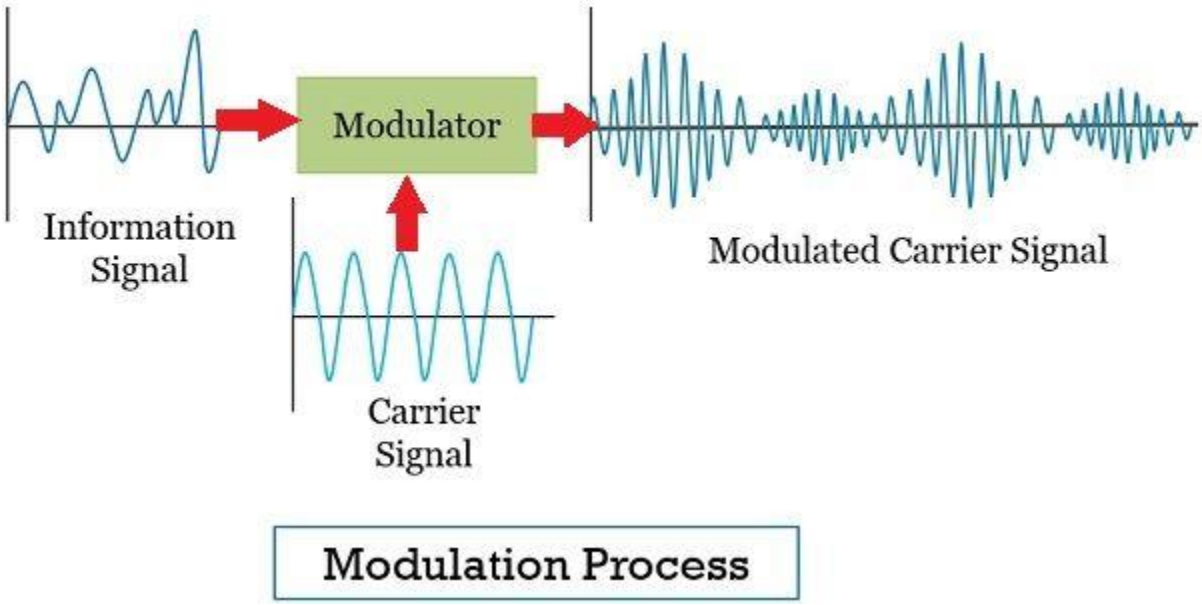


Fig.1: Amplitude modulation waveforms



The process of extracting the information bearing signal from the modulated bandpass signal is known as demodulation or detection. To recover the low frequency baseband signal, the received signal is first rectified and then filtered as depicted in Figure 2.

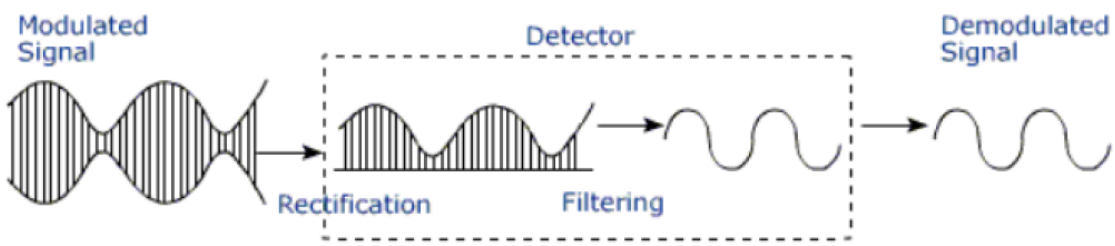


Fig. 2: Illustration of an amplitude demodulation

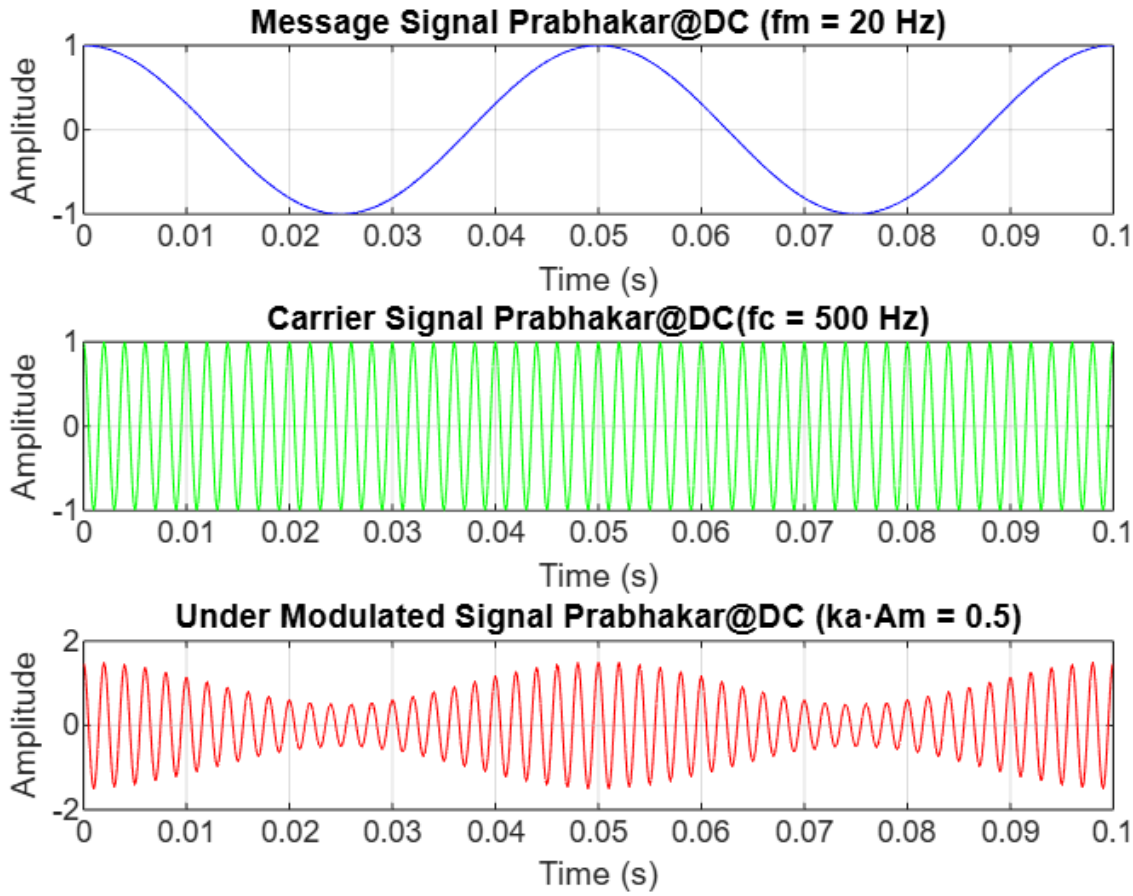
modulation index (μ = 0.5)

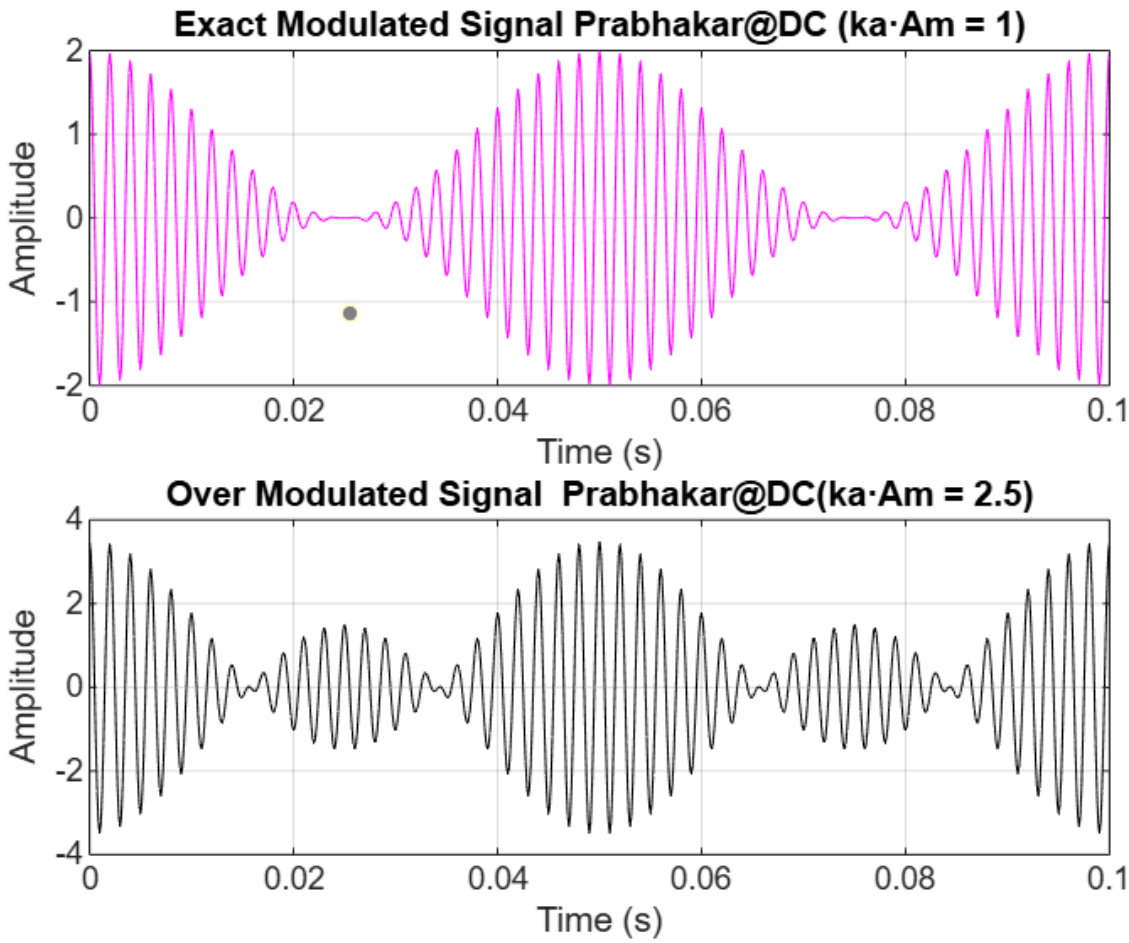
- μ < 1 → **Under-modulated** (safe, no distortion)
- μ = 1 → **100% modulation**
- μ > 1 → **Over-modulated** (distorted envelope, not desired)

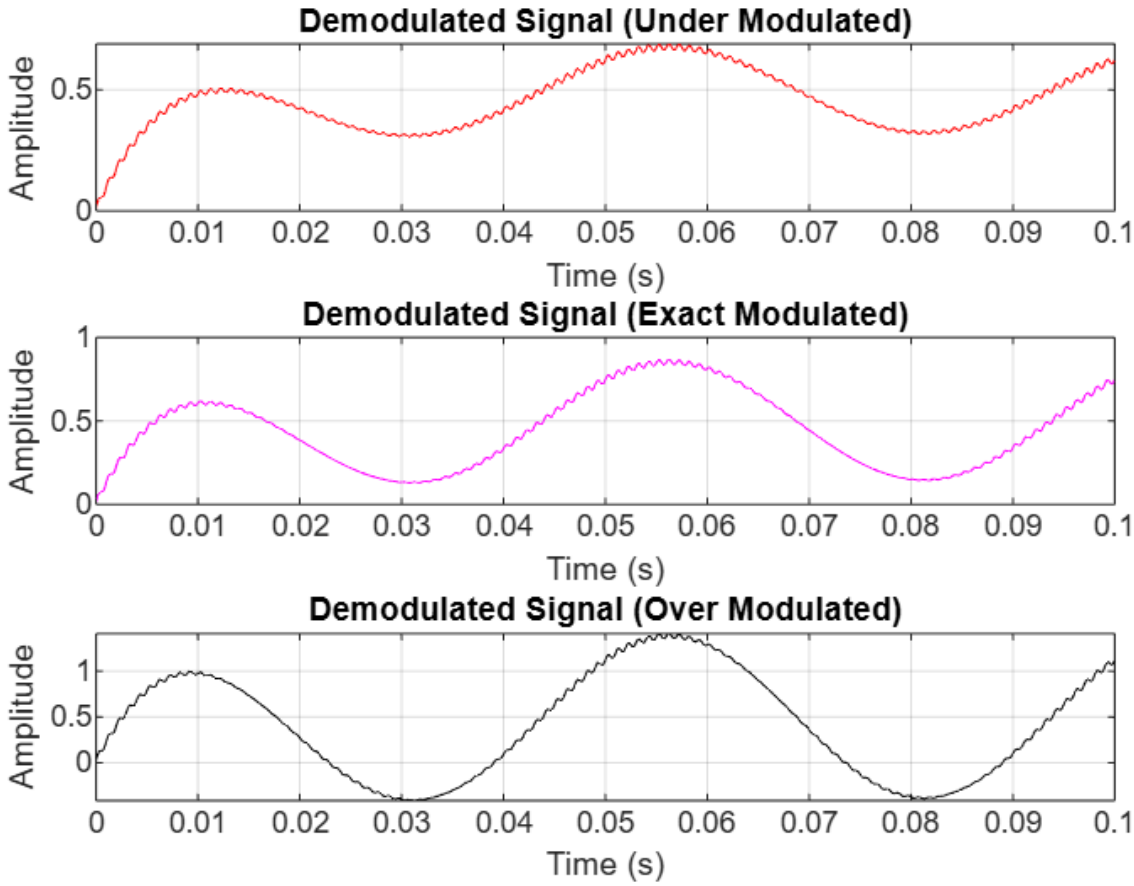
So **your AM system is 50% modulated**, which is a good condition.

```
clc;
clear;
close all;

% Parameters
fs = 8000;    % Sampling frequency
fm = 20;     % Frequency of message signal (Hz)
fc = 500;
Ac = 1;       % Amplitude of message signal
t = [0:0.1*fs]/fs; % Time vector (1 second)

%% ------------------ Figure 1: Message, Carrier, Under-Modulation ------------------
figure(1);
Am = 1;
ka = 0.5; % amplitude sensitivity .
u = ka * Am;

% Message signal
m = Am * cos(2*pi*fm*t);
subplot(3,1,1);
plot(t, m, 'b');
title('Message Signal (fm = 20 Hz)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Carrier signal
c = Ac * cos(2*pi*fc*t);
subplot(3,1,2);
plot(t, c, 'g');
title('Carrier Signal (fc = 500 Hz)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Under Modulated Signal (ka·Am = 0.5)
s1 = Ac * (1 + u * cos(2*pi*fm*t)) .* c;
subplot(3,1,3);
plot(t, s1, 'r');
title('Under Modulated Signal (ka·Am = 0.5)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

%% ------------------ Figure 2: Exact and Over Modulated ------------------
figure(2);

% Exact Modulated Signal (ka·Am = 1)
Am = 2; ka = 0.5; u = ka * Am;
s2 = Ac * (1 + u * cos(2*pi*fm*t)) .* c;
subplot(2,1,1);
plot(t, s2, 'm');
title('Exact Modulated Signal (ka·Am = 1)');
xlabel('Time (s)');
ylabel('Amplitude');
```

```
grid on;

% Over Modulated Signal (ka·Am = 2.5)
Am = 5; ka = 0.5; u = ka * Am;
s3 = Ac * (1 + u * cos(2*pi*fm*t)) .* c;
subplot(2,1,2);
plot(t, s3, 'k');
title('Over Modulated Signal (ka·Am = 2.5)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

%% ------------------ Figure 3: Demodulated Signals ------------------
figure(3);

% Low-pass filter
[b, a] = butter(1, fm/(fs/2)); % 1st order Butterworth LPF

% Demodulate Under Modulated
r1 = s1 .* c;
mr1 = filter(b, a, r1);
subplot(3,1,1);
plot(t, mr1, 'r');
title('Demodulated Signal (Under Modulated)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Demodulate Exact Modulated
r2 = s2 .* c;
mr2 = filter(b, a, r2);
subplot(3,1,2);
plot(t, mr2, 'm');
title('Demodulated Signal (Exact Modulated)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Demodulate Over Modulated
r3 = s3 .* c;
mr3 = filter(b, a, r3);
subplot(3,1,3);
plot(t, mr3, 'k');
title('Demodulated Signal (Over Modulated)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
```


Message Signal Prabhakar@DC (fm = 20 Hz)
Amplitude
1
0
-1
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1
Time (s)
Carrier Signal Prabhakar@DC(fc = 500 Hz)
Amplitude
1
0
-1
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1
Time (s)
Under Modulated Signal Prabhakar@DC (ka·Am = 0.5)
Amplitude
2
0
-2
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.1
Time (s)


Exact Modulated Signal Prabhakar@DC (ka·Am = 1)
Amplitude
Time (s)
Over Modulated Signal Prabhakar@DC(ka·Am = 2.5)
Amplitude
Time (s)



Discussion and conclusion: